

SEMINAR REFERENCE SHEET

FIRST-ORDER SCALAR LINEAR D.E.

$$x'(t) + a(t)x = f(t)$$

INTEGRATING FACTOR METHOD:

$$\rightarrow \mu(t) = e^{\int a(t) dt}$$

$$\rightarrow x'(t) + a(t)x = f(t) \mid \cdot \mu(t)$$

$$\Rightarrow \dots \int$$

$$\Rightarrow x(t) = \dots$$

LAGRANGE METHOD:

$$(1) x'(t) + a(t)x = f(t)$$

$$\rightarrow x'(t) + a(t)x = 0$$

$$\rightarrow x_h = C \cdot e^{-A(t)}$$

$$A(t) = \int a(t) dt$$

$$\rightarrow x_p(t) = \Phi(t) \cdot e^{-A(t)}$$

$$\rightarrow x_p'(t) = \dots$$

$$\rightarrow \text{substitute in (1)}$$

$$\rightarrow \Phi'(t) \cdot e^{-A(t)} = f(t)$$

$$\rightarrow \Phi'(t) = f(t) \cdot e^{A(t)} \int$$

$$\Rightarrow \Phi(t)$$

$$\Rightarrow x_p(t) = \dots$$

$$x(t) = x_h(t) + x_p(t)$$

HASTERS FORMULA:

$$x(t) = e^{-\int a(t) dt} \left(C + \int e^{\int a(t) dt} \cdot f(t) dt \right)$$

η^* equil. $\begin{cases} \text{unstable} \\ \text{stable} \\ \text{a. stable (stable + attractor)} \end{cases}$

η^* hyperbolic $\Leftrightarrow \operatorname{Re}(\lambda_1) \neq 0, \operatorname{Re}(\lambda_2) \neq 0 \dots$

origin: repeller/attractor $\Rightarrow \exists$ first integral.

$$x'(t) = \dot{x}(t) = \frac{dx}{dt}$$

Uncoupled linear systems with a center.

ELLIPSE $\rightarrow$ linear planar system with purely imaginary eigenvalues and diagonalizable matrix

Triangular matrix (ex: $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$) $\Rightarrow$ eigenvalues are on main diag $\Rightarrow 1, 2$ are eigenvals. $\Rightarrow 1 \neq 2$ distinct eigenvals $\Rightarrow A$ is diagonalizable.

Check diag: find eigenvals $\rightarrow$ find eigenvectors $\rightarrow$ count indep. eigenvectors $\rightarrow$ enough? $\Rightarrow$ diag. $\rightarrow$ not enough? $\Rightarrow$ not diag.

(PLANAR D.S.)

FIRST-ORDER SCALAR AUTONOMOUS D.E.

$$\dot{x} = f(x), f \in C^1(\mathbb{R})$$

$$\text{SCALAR ODE: } x(0) = \eta$$

SYSTEM in $\mathbb{R}^2$:

$$(x(0), y(0)) = (m_1, m_2)$$

$$\Rightarrow \varphi(t, \eta)$$

$$x(0) = \eta$$

EQUILIBRIUM: $f(\eta^*) = 0$.

$\exists!$: $x(0) = \eta^* \Rightarrow \exists!$ maximal sol.

FLOW: solution of IVP $\Rightarrow x(t) = \varphi(t, \eta)$

"the position at time t is the result of starting at η and letting the system run for t seconds".

$$\varphi(0, \eta) = \eta$$

$$\varphi(t, 0) = 0 \Leftrightarrow 0 \text{ equilibrium}$$

$$\varphi(t+s, \eta) = \varphi(t, \varphi(s, \eta))$$

D - open subset of $\mathbb{R}^2$ and φ is a C^1 function.

$$\varphi: D = \{(t, \eta) \in \mathbb{R}^2 : t \in I_\eta\} \rightarrow \mathbb{R}$$

$$\text{ORBIT: } \gamma_\eta = \{\varphi(t, \eta) : t \in I_\eta\} \subset \mathbb{R}$$

"all states visited by the solution".

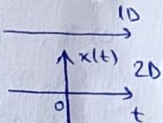
PHASE PORTRAIT of $\dot{x} = f(x)$ (for equations)

1) find equil: $f(x) = 0$

2) find sign of $f(x)$ on each orbit $\begin{cases} + \Rightarrow \text{arrow } > \\ - \Rightarrow \text{arrow } < \end{cases}$

3) draw line / graph

$\Rightarrow$ deduce attractor / repeller from arrows (stable) (unstable)



ATTRACTOR η^* is attractor when $\exists \epsilon > 0$ s.t. $|\eta - \eta^*| < \epsilon$, (equil.) we have $\lim_{t \rightarrow \infty} \varphi(t, \eta) = \eta^*$.

Basin of Attraction: $A_{\eta^*} = \{\eta \in \mathbb{R} : \lim_{t \rightarrow \infty} \varphi(t, \eta) = \eta^*\}$.

REPELLER: η^* is repeller when $\exists \epsilon > 0$ s.t. $|\eta - \eta^*| < \epsilon$, (equil.) we have $\lim_{t \rightarrow -\infty} \varphi(t, \eta) = \eta^*$.

LIN. METHOD: (for higher dimensions) (without arrows)

$f'(\eta^*) < 0 \Rightarrow \eta^*$ attractor $\Rightarrow$ stable (for systems)

$f'(\eta^*) > 0 \Rightarrow \eta^*$ repeller $\Rightarrow$ unstable

$f'(\eta^*) = 0 \Rightarrow$ fails $\Rightarrow$ use sign analysis...

repeller: $A_{\eta^*} = \{\eta \in \mathbb{R}^n : \lim_{t \rightarrow -\infty} \varphi(t, \eta) = \eta^*\}$

Diagonalizable matrix: $A = P \cdot D \cdot P^{-1}$ (non-)

D = diag. matrix of eigenvalues

P = matrix whose columns are eigenvectors of A .

A diag. $\Leftrightarrow A$ has enough lin. ind. eigenvectors

SECOND ORDER LDE w CC

$$ax'' + bx' + cx = 0$$

ALGORITHM FOR DET. THE GEN. SOL.
FOR THE m 'th order LHDE w CC.

$$x^{(m)} + a_{m-1}(t)x^{(m-1)} + \dots + a_1(t)x' + a_0(t)x = f(t), (t) \in I.$$

- $f(t) = 0 \Rightarrow$ linear, homog.
- $f(t) \neq 0 \Rightarrow$ linear, non-homog.

$\rightarrow$ Characteristic equation:

$$r^m + a_{m-1}r^{m-1} + \dots + a_1r + a_0 = 0$$

$\rightarrow$ Det. roots of char. eq.

$\rightarrow$ To each root attach a function:

a) $r_1, \dots, r_m \in \mathbb{R}; r_1 \neq \dots \neq r_m$

$$x_1(t) = e^{r_1 t}, x_2(t) = e^{r_2 t}, \dots$$

b) $r_1, \dots, r_m \in \mathbb{R}; r_1 = \dots = r_m$

$$x_1(t) = e^{r_1 t}$$

$$x_2(t) = t \cdot e^{r_1 t}$$

$$x_m(t) = t^{m-1} \cdot e^{r_1 t}$$

c) $r_1, \dots, r_m \in \mathbb{C}; r_1, \dots, r_m = \alpha \pm i\beta$

$$\begin{cases} x_1(t) = e^{\alpha t} \cdot \cos \beta t \\ x_2(t) = e^{\alpha t} \cdot \sin \beta t \end{cases}$$

$$\begin{cases} x_3(t) = t \cdot e^{\alpha t} \cdot \cos \beta t \\ x_4(t) = t \cdot e^{\alpha t} \cdot \sin \beta t \end{cases}$$

$\rightarrow$ Write general solution:

$$x(t) = c_1 \cdot x_1(t) + c_2 \cdot x_2(t) + \dots + c_m \cdot x_m(t)$$

$$c_1, \dots, c_m \in \mathbb{R}.$$

m^* hyperbolic when $\text{Re}(\lambda) \neq 0$, (b) $\lambda \dots$

SEPARATION OF VARIABLES METHOD

PLANAR AUT. SYSTEMS:

$$\begin{cases} \dot{x} = f(x, y) \\ \dot{y} = g(x, y) \end{cases}$$

$$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}} = \frac{g(x, y)}{f(x, y)}$$

$$(\Rightarrow) f(x, y)dy = g(x, y)dx = 0$$

LINEAR PLANAR SYSTEMS:

$$\begin{cases} \dot{x} = ax + by \\ \dot{y} = cx + dy \end{cases}$$

$$\frac{dy}{dx} = \frac{cx + dy}{ax + by}$$

$$(\Rightarrow) (ax + by)dy = (cx + dy)dx$$

FOR D.E:

$$\dot{x} = f(x)$$

$$\frac{dx}{f(x)} = dt \quad | \int$$

$$\int \frac{dx}{f(x)} = \int dt = t + C$$

ex: $x' = 3x - x^2$

$$\frac{dx}{dt} = x(3-x)$$

$$\frac{dx}{x(3-x)} = dt \quad | \int$$

$$\Rightarrow x = \dots$$

(LINEAR PLANAR SYSTEMS)

FIRST-ORDER SCALAR LINEAR HOMOG. SYSTEMS w CC

$$\dot{x} = Ax, A \in M_2(\mathbb{R}), \det A \neq 0$$

EQUILIBRIUM: $x^* = 0$.

ALGORITHM FOR DET. THE GEN. SOL. & FLOW

$\rightarrow$ Eigenvalues: $\det(A - \lambda I_m) = 0$

$\rightarrow$ Eigenvectors: $(A - \lambda I_m)v = 0$ for each eigenvalue.

$\rightarrow$ Build solutions:

a) $\lambda_1, \lambda_2 \in \mathbb{R}$ and either

(a. stable) $\lambda_1 \leq \lambda_2 < 0$ or

$0 < \lambda_1 \leq \lambda_2$ (a. unstable) global attr.

no oscillations $\leftarrow$ (NODE)

(lines)

$\alpha < 0$: toward 0

$\alpha > 0$: away from 0

$\sim$ hyperbolas

(5)

(2)

closed orbits $\leftarrow$ (CENTER)

a. stable $\Rightarrow$ stable. (ellipses/circles)

b) $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\lambda_1 < 0 < \lambda_2$ $\leftarrow$ (SADDLE) $\rightarrow$ unstable. (Th. 2) always

c) $\lambda_{1,2} = \alpha \pm i\beta, \alpha \neq 0, \beta \neq 0$

Spirals

$\leftarrow$ (FOCUS) $\rightarrow$ stable ($\alpha < 0$) asymptotically

unstable ($\alpha > 0$) asymptotically

d) $\lambda_{1,2} = \pm i\beta, \beta \neq 0$

stable. (Th. 2) always.

(not a. stable)

$$\det A = \lambda_1 \cdot \lambda_2 \text{ (general)}$$

$\bullet \det A \neq 0 (\Rightarrow) 0 \in \mathbb{R}^2$ only equilibrium.

$(\Rightarrow) \lambda_1 \neq 0$ and $\lambda_2 \neq 0$.

GLOBAL ATTRACTOR:

$0 \in \mathbb{R}^2$ global attractor if $\begin{cases} \text{Re}(\lambda_1) < 0 \\ \text{Re}(\lambda_2) < 0 \end{cases}$

(equil.) $\Rightarrow$ a. stable

GLOBAL REPELLER:

$0 \in \mathbb{R}^2$ global repeller if $\begin{cases} \text{Re}(\lambda_1) > 0 \\ \text{Re}(\lambda_2) > 0 \end{cases}$

(equil.) $\Rightarrow$ unstable

$x' \rightarrow$ left/right
 $y' \rightarrow$ up/down

$$x(t) = x_h(t) + x_p(t)$$

$\rightarrow$ char. eq. for homog. eq.

$\Rightarrow r_1, r_2$

$$\Rightarrow x_h(t) = \dots$$

$\rightarrow$ form of $x_p(t) \rightarrow x'_p(t) \dots$

$\rightarrow$ substitute in initial eq.

$\rightarrow$ equate coefficients

$\rightarrow$ back in $x_p(t) = \dots$

$\rightarrow x(t) = \dots$

Reduce 2x2 lin. system to one second order...

$$\begin{cases} \ddot{x} = 4y & \ddot{y} = -4x \Rightarrow x'' = -4x \dots \\ \dot{y} = -x \end{cases}$$